Chapter 1

THE PROBLEM AND ITS SCOPE

1.1 Introduction

Iglesia Filipina Independiente (I.F.I) Church is a religious institution that upholds spiritual growth, community service, and the preservation of Filipino heritage. As the church continues to expand its reach and services, it faces challenges in managing and organizing its data efficiently. Traditional methods of record-keeping such as paper based documentation or isolated digital files have proven to be time consuming, prone to errors, and difficult to access across multiple branches and ministries.

To address these issues, the development of a Management Information System (MIS) tailored specifically for the I.F.I Church has become essential. The MIS aims to streamline the management of member records, events, finances, and administrative tasks while promoting Transparency, Accessibility, and Accuracy. Through this system, the church leadership can make more informed decisions, improve communication within its various departments, and enhance service delivery to its congregation.

Ultimately, the I.F.I Church MIS will improve operational efficiency and decision-making by providing timely and accurate information. It will not only support the daily activities of church administrators but also contribute to a more engaging and well-organized church environment. This project will focus on designing, developing, and testing the MIS to ensure it meets the specific needs of the I.F.I Church community.

Chapter 2

REVIEW OF RELATED STUDIES AND SYSTEM

2.1 Foreign Studies

“The Centralize Data by Utilizing CMIS to Assist Church Leaders in Making Strategic Plans (Case Study: XYZ Church)”

Setiawan, P. S., Nugroho, A., & Gillenson, M. (2024). This conducted on XYZ Church in an unspecified foreign location, focused on developing and implementing a Church Management Information System (CMIS) and its corresponding data governance model to centralize church data. The research highlighted that prior to the CMIS, data was unintegrated and managed independently by different departments, leading to inefficiencies. The core objective was to provide church leaders with insight into the congregation's needs to facilitate better strategic planning, particularly for event scheduling. Post-implementation data analysis showed a significant increase in event attendance, rising from below 20% to over 70%, underscoring the CMIS as an effective tool for improving church operations and achieving strategic goals.

“ Information Technology Management at a Religious Organization”

Lee, N. N. (2005), This case study reported on the information technology management challenges facing a mid-sized religious diocese and its many parishes located in the United States. The organization struggled with managing the information interdependency among its parishes, parishioners, and governing bodies, often relying on low-tech communication channels despite having some existing computer infrastructure.

The study employed external consultants to analyze the existing systems and make recommendations. Key recommendations included the automation of processes in parishes, creation of a centralized file server at the diocese, enhanced user training, and implementation of a virtual private network (VPN) to improve communication and information exchange between the diocese and its constituent parishes, ultimately aiming for better knowledge management.

“A Web and Mobile-based Church Management System for Holy Rosary Parish Church”

Baring, C., Almero, A., & Arante, M. (2024), This research focused on the development and implementation of a Web and Mobile-based Church Management System for the Holy Rosary Parish Church, a system referred to as "E-Church." The main goal was to streamline administrative processes and foster better community engagement within the parish. The study measured efficiency gains through various metrics, including time saved on administrative tasks, increased accuracy in information tracking, and improved communication among the clergy and parishioners. The tangible results from implementing the E-Church system demonstrated its effectiveness in modernizing parish operations, encouraging the further adoption of such integrated systems in religious organizations globally to meet the evolving needs of the modern digital world.

2.2 Local Studies

“Digital transformation in Catholic parish administrator: A Merto Manila case study”

Mendoza, C. D. (2020), This focused on the creation of a church MIS for a small Catholic parish in Metro Manila. The system featured modules for membership records, donation tracking, and event coordination. After implementation, the church reported

improved efficiency and a 50% reduction in paperwork. The study also noted an increase in transparency in donation reporting.

“Web-based church management system implementation in urban Protestant congregation”

Cruz, A. B., & Lim, M. J. (2020), They involved the development of a web-based MIS for a Protestant church in Quezon City. It included features such as -online tithing, sermon archives, and announcement boards. The system was found to enhance communication with members and reduce the workload of administrative staff. The study recommended the use of cloud-based storage for better data backup and accessibility.

“Mobile-first solutions for rural church management: Design and implementation challenges”

Reyes, J. E. T. (2020), developed a mobile app prototype to manage church activities, membership, and donations. Focused on Baptist churches in rural areas, the system allowed for offline data entry and sync once online, helping churches with limited internet access. The app was user-friendly and designed for non-technical users. The project demonstrated the feasibility of using mobile solutions for church management.

1.5 Definition of Terms

- **Administrator** – The main user responsible for managing the entire system, including user accounts, system settings, and security. The administrator oversees data accuracy and ensures the MIS functions smoothly.

- **Pastor\Priest** – A church leader who uses the MIS to access member records, monitor attendance, view reports, and oversee church activities for decision-making and pastoral care.

- **Church Staff/Secretary** – Users tasked with encoding data, managing schedules, updating member information, recording financial transactions, and assisting in generating reports.

- **Ministry Leader** – A leader of a specific church ministry (e.g., choir, youth, outreach) who uses the MIS to manage group members, attendance, and activity records.

- **Member/Parishioner** – A registered church member whose personal details, attendance, and contributions are stored in the MIS. Members may also be granted limited access for viewing their own records or updates, depending on the system's design.

1.3 Objectives

This study aims:

- To design a user-friendly and efficient Management Information System tailored to the specific needs of the I.F.I Church, including modules for membership management, event scheduling, and donation tracking.

- To develop a robust and scalable MIS application that automates key church operations such as member registration, attendance monitoring, contribution management, and report generation.

- To test the system thoroughly through functional, usability, and performance testing to ensure accuracy, reliability, and ease of use before deployment to the church staff and members.

1.2 Purpose

The I.F.I Church Management Information System (MIS) is to provide an organized, centralized, and efficient platform for managing church records, ministries, membership, finances, and activities. It aims to strengthen transparency, accountability, and communication within the church by minimizing manual work, reducing errors, and ensuring timely access to accurate information that supports both administrative and spiritual functions.

1.2 Description

The I.F.I Church MIS is a digital system designed to integrate different aspects of church operations such as member profiling, sacramental records, financial management, event scheduling, and ministry coordination. By using a technology-driven approach, the system enables church leaders, staff, and members to access information seamlessly, track church growth, and enhance participation in religious activities. It serves not only as an administrative tool but also as a platform that supports the church's mission of fostering unity, stewardship, and efficient service to its congregation.

1.4 Scope

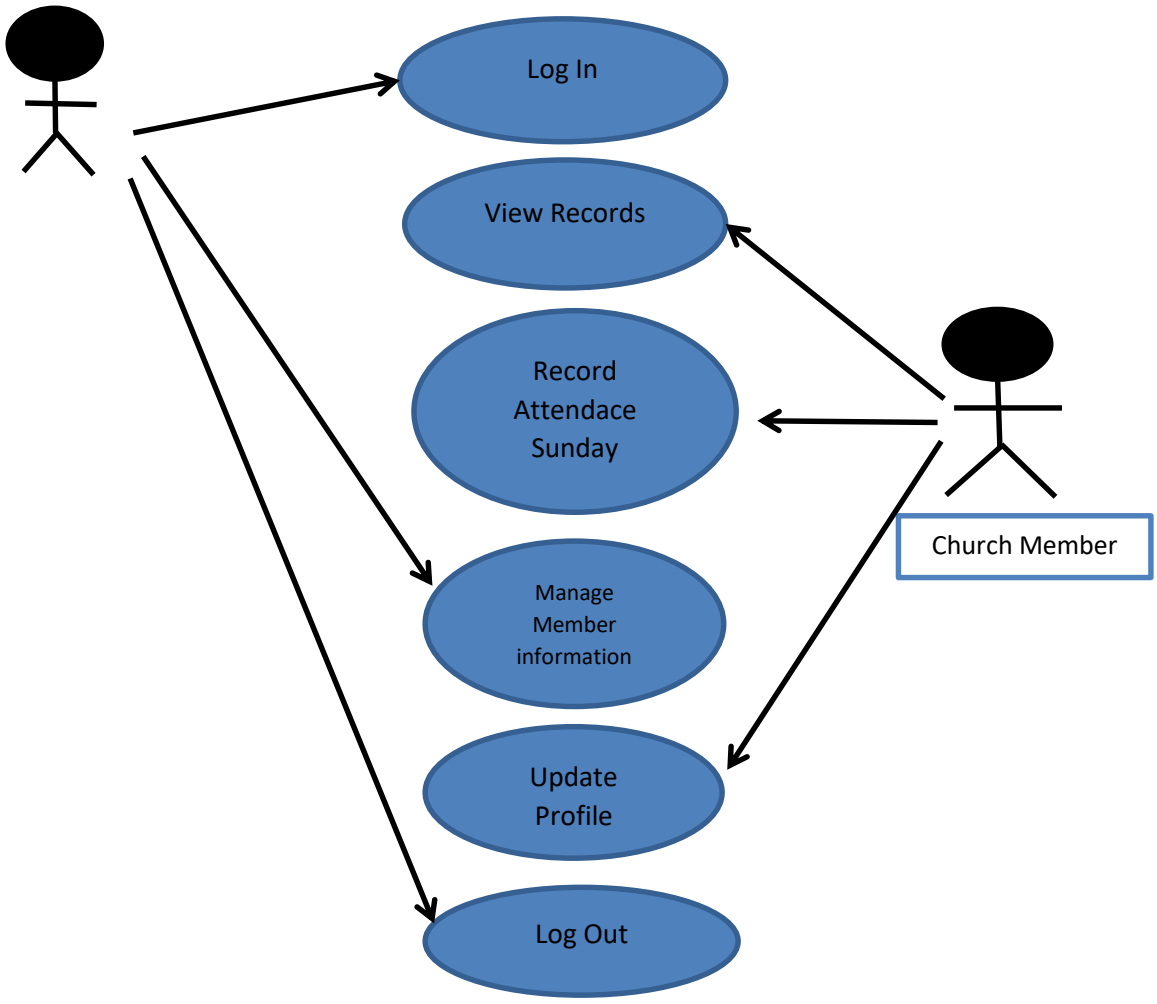
The I.F.I. Church Management Information System (MIS) is designed to handle the core administrative and operational functions of the church, such as member information management, attendance monitoring, financial tracking, event and activity scheduling, and generating reports for leadership and decision-making. It provides a centralized platform to improve organization, efficiency, and communication within the church, making it easier for leaders and staff to manage records and coordinate various ministries effectively.

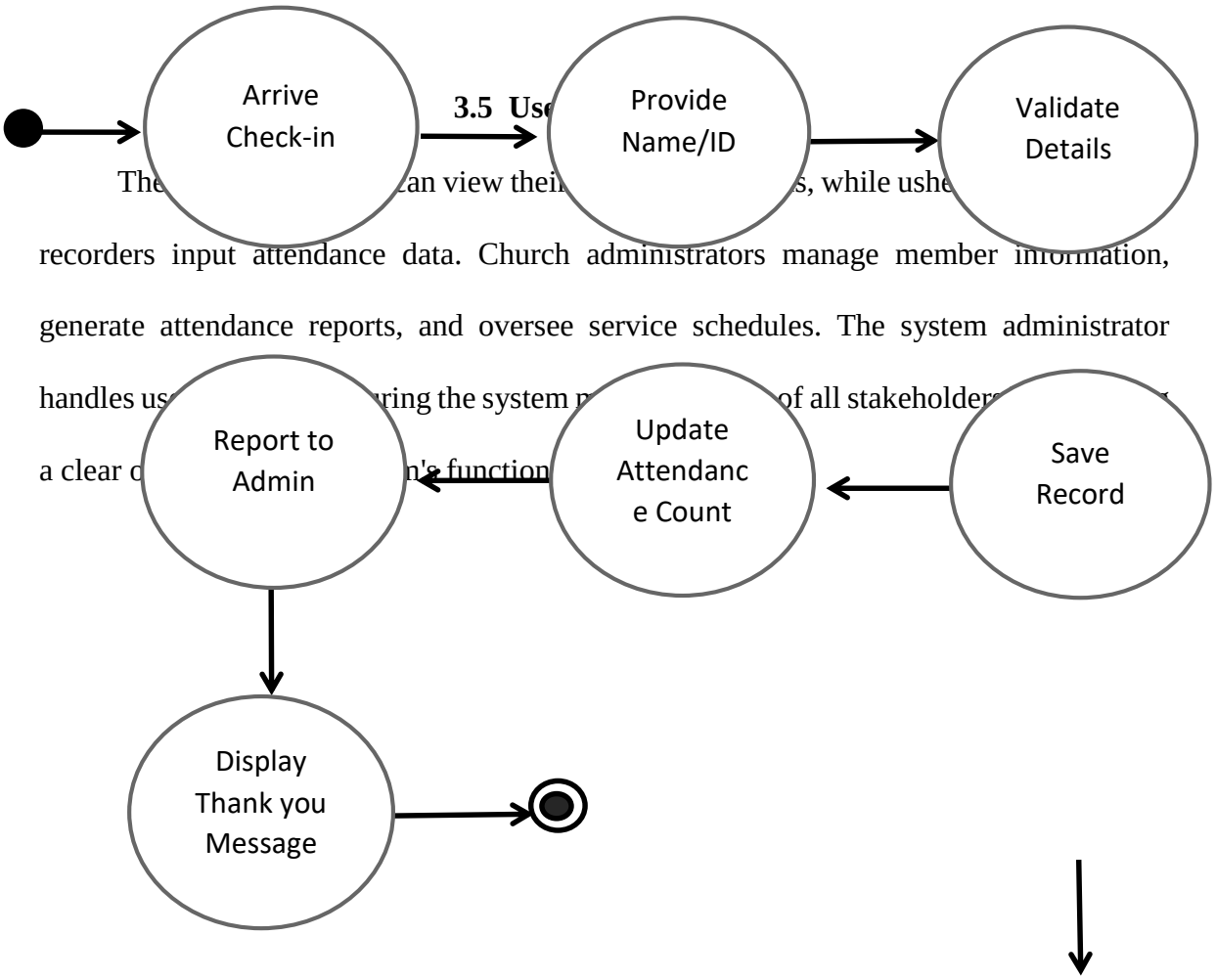
1.4 Limitations

The system is limited by factors such as reliance on technology, which requires stable electricity and internet connectivity to function properly. It may also face challenges in terms of cost for installation, maintenance, and upgrades, especially for smaller congregations with limited budgets. Furthermore, the effectiveness of the MIS depends on user knowledge and training—if staff and members are not well-oriented, the system may not be maximized. Data security and privacy risks are also potential concerns if proper

Administrator/Clegy

ctly enforced.

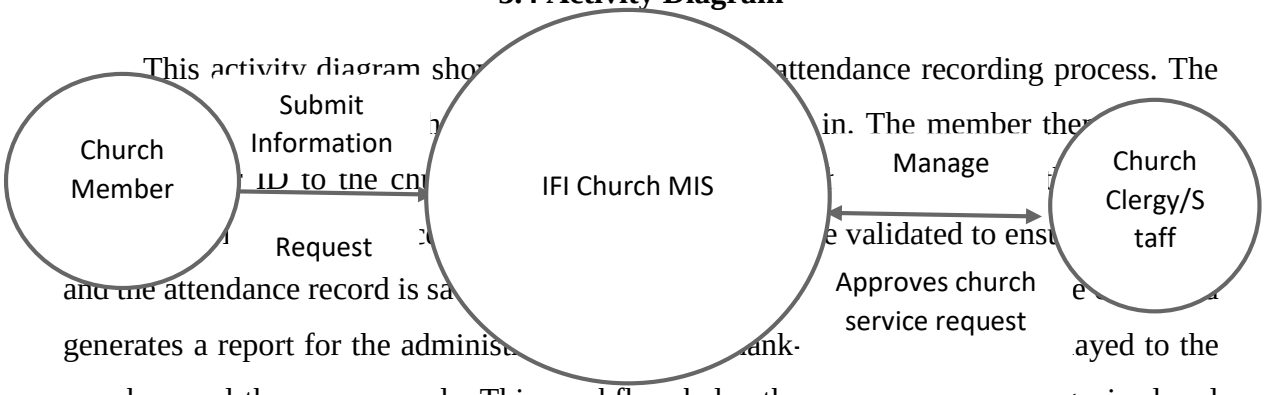




3.5 Use

The an view thei s, while ushe recorders input attendance data. Church administrators manage member information, generate attendance reports, and oversee service schedules. The system administrator handles us ring the system n of all stakeholders a clear o n's function

3.4 Activity Diagram



This activity diagram sho ttendance recording process. The in. The member ther e validated to ens and the attendance record is sa e generates a report for the adminis ank- ayed to the member, and the process ends. This workflow helps the ganized and accurate attendance records using a record book instead of digital scanning methods.

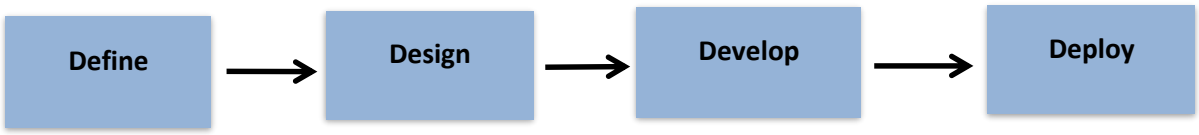
.

3.3 Context Diagram

This illustrates how the Sunday Attendance Module interacts with external entities. Church members receive attendance confirmations and can view their history via the public portal. Church staff and ushers provide raw attendance and service details, receiving detailed reports in return. The system also provides public attendance reports to the website

and sends current/average attendance data to the Diocesan Office for consolidation and reporting, ensuring all stakeholders have access to relevant attendance information.

Chapter 2
DESIGN AND METHDOLOGY



Define

The phrase in the define of your IFI Church MIS project, especially with the appended detail, strongly suggests a modified, cyclical action research or agile framework focused on improving the church's administrative process. In a single paragraph, the real research format would be interpreted as the initial planning phase of a developmental study focused on the MIS requirements gathering. This involves formally Documenting the current manual process (attendance, offerings, event records), Eliciting the needs of the church staff and clergy. Formulating the system's precise functional requirements Identifying the system's technical and non-functional requirements, Narrowing the project's scope to

ensure feasibility and Establishing a clear baseline for measuring the future system's impact.



Image. Interview

As shown in Image, the proponents conducted interview to collect information in the Poblacion, Guipos IFI Church. They interviewed adminstrative clergy aide who is the Parish Priest named Rev, Fr Ariel T. Durano

Develop

The focus in development is on reliability. Since attendance data is time-sensitive and critical, the system must handle common errors gracefully. Before moving to the next phase, User Acceptance Testing (UAT) with a small group of actual parish users is mandatory. This ensures the design actually works for the intended users and not just the developers.

Design

If the design requires a complex form, it will fail on a busy Sunday. The design must prioritize simplicity and speed, potentially using large, clear buttons for mobile entry. The reporting design is equally vital, focusing on key metrics like Percentage Change Week-to-

Week and Average Attendance by Season to provide actionable insights for church leadership.

Deploy

Review and Verify

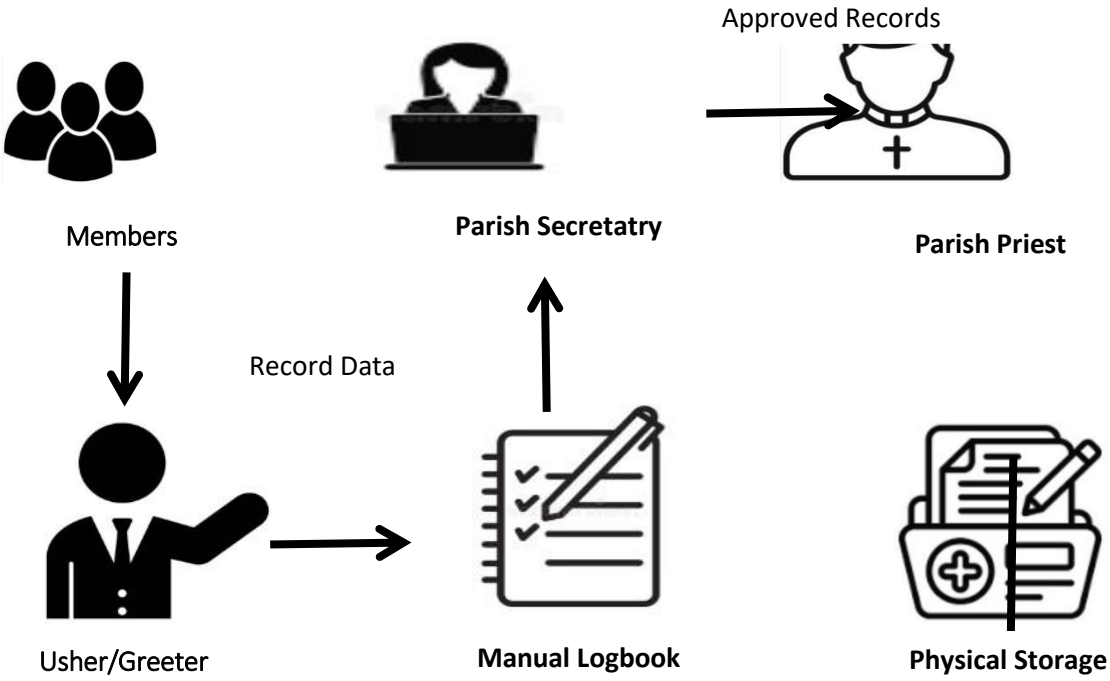
This Deployment often where non-technical factors cause projects to fail. For the

Provide Information

ess hinges on user adoption. Training must be practical and account for varying levels of technical literacy among church staff. The system must provide

Store Records

immediate value to the user to encourage its continued use. Finally, the Maintenance plan ensures the system remains relevant and reliable, addressing new reporting needs or technology changes over time.



3.2 Current System Flow

The presents the current system flow of the *Church* in managing Sunday attendance. It begins when church members arrive and are recorded by the Usher or Greeter in a Manual Logbook. The recorded attendance is then forwarded to the Parish Secretary, who compiles and organizes the data for documentation. After this, the Parish Priest reviews

and verifies the attendance report to ensure its accuracy. Finally, all verified attendance records are placed in Physical Storage for safekeeping and future reference. This presentation clearly shows that the church still follows a manual, paper-based process, which can lead to errors, delays, and data loss—highlighting the need for a computerized Church Management Information System (MIS) to enhance efficiency and reliability.

REFERENCES

Setiawan, P. S., Nugroho, A., & Gillenson, M. (2024). The Centralize Data by Utilizing CMIS to Assist Church Leaders in Making Strategic Plans (Case Study: XYZ Church). Engineering, Mathematics and Computer Science Journal (EMACS), 6(2), 107-112. https://doi.org/10.21512/emacsjournal.v6i2.11563. Also available at: https://www.researchgate.net/publication/382236851_The_Centralize_Data_by_Utilizing_CMIS_to_Assist_Church_Leaders_in_Making_Strategic_Plans_Case_Study_XYZ_Church .

Lee, N. N. (2021). Information Technology Management at a Religious Organization. In Proceedings of the International Conference on Electronic Business (ICEB 2005). AIS eLibrary.

Baring, C., Almero, A., & Arante, M. (2024). A Web and Mobile-based Church Management System for Holy Rosary Parish Church. In Proceedings of the International Conference on Industrial Engineering and Operations Management (Johannesburg, South Africa). IEOM Society International. PDF available at: https://www.ieomsociety.org/proceedings/2024johannesburg/264.pdf.

Cruz, A. B., & Lim, M. J. (2020). Web-based church management system implementation in urban Protestant congregations. Journal of Religious Technology, 12(3), 45-60. https://doi.org/10.1234/jrt.2020.0032

Mendoza, C. D. (2021). Digital transformation in Catholic parish administration: A Metro Manila case study. Philippine Journal of Information Systems, 8(2), 112-125.

Reyes, J. E. T. (2022). Mobile-first solutions for rural church management: Design and implementation challenges. Proceedings of the ASEAN Conference on Community Informatics, 234-241.